Hanall Sung, Ph.D.

University of Tennessee, Knoxville
Educational Data Science
Suite 514, 1126 Volunteer Blvd, Knoxville, TN 37916 USA
hanallsung@utk.edu, +1-608-960-3769

AREAS OF RESEARCH INTERESTS

(Multimodal) Learning Analytics, Human-AI Interaction, Emerging Learning Technologies (AR/VR), Computer-Supported Collaborative Learning, STEM Education

PROFESSIONAL APPOINTMENT

Assistant Professor (Tenure-track) of Educational Data Science
University of Tennessee, Knoxville
College of Education, Health, and Human Sciences
Department of Theory & Practice in Teacher Education

Postdoctoral Fellow in Learning Sciences and Psychological Studies
University of North Carolina at Chapel Hill

EDUCATION

Ph.D., Educational Psychology (Learning Sciences)
University of Wisconsin–Madison

M.S., Educational Psychology (Learning Sciences)
University of Wisconsin–Madison

M.A. in Educational Technology
Ewha Womans University, South Korea

B.A. in Educational Technology
Ewha Womans University, South Korea

AWARDS • HONORS • SCHOLARSHIP

Oct 2022	**Doctoral Consortium Scholarship** 4th International Conference on Quantitative Ethnography (ICQE)
Jun 2022	**Patricia and Michael Busk Travel Scholarship** School of Education, University of Wisconsin–Madison
Jun 2022	**CSCL Naomi Miyake Best Student Paper Award**

	15th International Conference on Computer-Supported Collaborative Learning (CSCL)
Feb 2022	**Student Research Grants Competition-Conference Presentation Award** University of Wisconsin–Madison
Sep 2021– May 2022	**Graduate School Fellowship** College of Education, University of Wisconsin–Madison
Mar 2021	**Wallace Student Scholarship** International Society of the Learning Sciences (ISLS)
Sep 2020	**Bridge to Success Scholarship** School of Education, University of Wisconsin–Madison
Apr 2019	**Student Research Grants Competition-Conference Presentation Award** University of Wisconsin–Madison
Sep 2017– May 2018	**Graduate School Fellowship** College of Education, University of Wisconsin–Madison
Sep 2016	**Best Master Thesis Award** College of Education, Ewha Womans University, South Korea
Feb 2015	**Highly Commended Poster Presentation Award** 3rd Dubai International Conference in Higher Education, United Arab Emirates
Sep 2014	**Gold Prize in Best Paper Award** E-learning Korea 2014

RESEARCH GRANTS

Extramural Grants

Funded and in progress (2024–2026)	**Artificially Intelligent Communication of Science** - *Funding source*: College For Social Sciences and Humanities, University Alliance Ruhr, Germany (€11,500) - *Role*: International partner (PIs: Valentina Nachtigall and Maximilian Krug; International partner: **Hanall Sung**)
Unfunded (2025–2029)	**Validating Learning Analytics with Student Voice by Aligning Traces, Text, and Interview Data about Self-Regulated STEM Learning** - *Funding source*: National Science Foundation (NSF) ($484,534) - *Role*: Co-PI (PI: Joshua Rosenberg; Co-PIs: **Hanall Sung**, Shalaunda Reeves, Alex Lishinski, and Gregory Peterson)

Intramural Grants

Funded and in progress (2025–2026)	**Development of the Computing, AI, and Data in Educational Systems (CADES) Center** - *Funding source*: College of Education, Health, & Human Sciences, University of Tennessee, Knoxville ($3,500) - *Role*: Organizers/Lead faculty affiliates (Co-lead: Emily Holtz, Joshua Rosenberg, Louis Rocconi; Organizers/Lead faculty affiliates: **Hanall Sung**, Kelly Boles, Sukanya Moudgalya, Rachel Wong, Fatima Zahra)
Funded and in progress (2024–2027)	**Designing AI-enhanced XR Learning Interventions for STEM Education** - *Funding source*: College of Education, Health, & Human Sciences, University of Tennessee, Knoxville ($107,408) - *Role*: PI (PI: **Hanall Sung**)
Completed (2024–2025)	**AI for Education: Developing Curricula, Planning for External Funding, and Connecting Faculty** - *Funding source*: AI Tennessee Initiative ($21,000) - *Role*: Co-PI (PI: Louis Rocconi; Co-PIs: **Hanall Sung,** Joshua Rosenberg, Rachel Wong, Joshua Fagan, Xiaopeng Zhao, Gregory Peterson, Kelly Boles)

PUBLICATIONS

Articles published in refereed journals

Published

* indicates corresponding author

Lee, H. H.*, **Sung, H.**, Celedón-Pattichis, S., & Pattichis, M. (2026). Toward an inclusive understanding of collaborative learning using MMLA: Exploring multimodal participation dynamics by gender and linguistic diversity. *Learning and Instruction, 102*(102317). https://doi.org/10.1016/j.learninstruc.2026.102317 [SSCI-indexed; 5-year IF: 6.4]

Sung, H.* & Nathan, M. J. (2025). Unraveling temporally entangled multimodal interactions: Investigating verbal and nonverbal contributions to collaborative construction of embodied math knowledge. *International Journal of Educational Technology in Higher Education, 22*(8). https://doi.org/10.1186/s41239-025-00504-6 [SSCI-indexed; 5-year IF: 15.5]

Na, H. & **Sung, H.*** (2025). Learn math through motion: A technology-enhanced embodied approach with augmented reality for geometry learning in K-12 classrooms. *Interactive Learning Environments*, *33*(6), 3744–3757. https://doi.org/10.1080/10494820.2025.2450663 [SSCI-indexed; 5-year IF: 6.2]

Sung, H.*, Bernacki, M. L., Greene, J. A., Yu, L., & Plumley, R. D. (2024). Beyond frequency: Using epistemic network analysis and multimodal traces to understand temporal dynamics of self-regulated learning. *Journal of Science Education and Technology*, *34,* 1110-1127. https://doi.org/10.1007/s10956-024-10164-2 [SSCI-indexed; 5-year IF: 5.1]

Sung, H.* & Nathan, M. J. (2024). Your body tells how you engage in collaboration: Machine-detected body movements as indicators of engagement in collaborative math knowledge building. *British Journal of Educational Technology*, *55*(5), 1950-1973. https://doi.org/10.1111/bjet.13473 [SSCI-indexed; 5-year IF: 8.1]

Sung, H.*, Rau, M. A., & Van Veen, B. D. (2024). Development of an intelligent tutoring system that assess internal visualization skills in engineering using multimodal triangulation. *IEEE Transactions on Learning Technologies*, 17, 1625-1638. https://doi.org/10.1109/TLT.2024.3396393 [SSCI- and SCIE-indexed; 5-year IF: 5.5]

Sung, H.* & Jo, I. (2018). Utilizing multimodal data to predict learning achievement: Behavioral log, psychophysiological response, and test anxiety. *Journal of Educational Technology, 34*(2), 287-308. https://doi.org/10.17232/KSET.34.2.287 [KCI-indexed; 5-year IF: 3.86]

Jo, I., Park, Y*., Yoon, M., & **Sung, H.** (2016). Evaluation of online log variables estimating learners' time management in Korean online learning context. *The International Review of Research in Open and Distance Learning, 17*(1), 195-213. https://doi.org/10.19173/irrodl.v17i1.2176 [SSCI-indexed; 5-year IF: 2.8]

Manuscripts submitted for publication

Lim, J.*, **Sung, H.**, & Zhu, Y. (under review). Preservice Teachers' Digital/AI Competence Development in GenAI-Supported Lesson Planning: Orchestrating SQD2 Strategies. *Computers & Education.* [SSCI-indexed; 5-year IF: 13.0]

Sung, H.* (under review). Human–AI collaboration in multimodal learning analytics: A framework for advancing the study of collaborative learning in the age of AI. *International Journal of Computer-Supported Collaborative Learning.* [SSCI-indexed; 5-year IF: 5.7]

Sung, H.*, Barro, M., Yu, L., & Plumley, R. D., Bernacki, M. L., Green, J. A. (under 1st round review). Adhered or Independent: How students' adherence to high-structure active learning shapes self-regulated learning processes and academic outcomes. *British Journal of Educational Technology.* [SSCI-indexed; 5-year IF: 8.1]

Krug, M.*, Nachtigall, V., & **Sung, H.** (under 1st round review). AI labels undermine trust in scientific presenters. *Journal of Broadcasting & Electronic Media*. [SSCI-indexed; 5-year IF: 3.8]

Sung, G.* & **Sung, H.** (under 2nd round review). Connecting blinks to constructs: A review-based validity evidence framework for multimodal learning analytics. *Review of Educational Research.* [SSCI-indexed; 5-year IF: 15.4]

In preparation

Sung, H.* (in progress). Human–AI hybrid intelligence in multimodal learning analytics: A framework applied to embodied learning.

Sung, H.* & Nathan, M. J. (in progress). Advancing multimodal assessment in mathematics: Online embodied professional learning for teacher noticing and attitudes toward gesture.

Book Chapter

Kim, J., Lee, H., Yoo, Y., **Sung, H.**, Jo, IH., Park, Y. (2015). Towards smart asynchronous discussion activity: Using social network analysis to investigate students' discussion patterns. In: Chen, G., Kumar, V., Kinshuk, Huang, R., Kong, S. (eds) *Emerging Issues in Smart Learning. Lecture Notes in Educational Technology.* Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-44188-6_50

Papers published in refereed conference proceedings

Sung, H., Rho, J., & Kim, T. (accepted). "I thought you were watching me": Mismatches between learner expectations and AI capabilities in embodied virtual reality learning. *The International Conference on Learning Sciences and Educational Innovation (ICOLSEI 2026)*, Seoul, Korea.

Nachtigall, V., Krug, M., & **Sung, H.** (accepted). Uncritical but overconfident? How learners engage with AI-generated images of scientists. *The 27th International Conference on AI in Education (AIED 2026)*, Seoul, Korea.

Sung, H. (accepted). Multimodal interactions with AI: Human–AI partnerships in learning, teaching, and research. *Proceedings of the 20th International Conference of the Learning Sciences - ICLS 2026*, Irvine, CA, United States.

Sung, H., Greene, J. A., Bernacki, M. L., Berro, M., Yu, L., & Plumley, R. D. (2026). Epistemic network analysis and multigroup modeling reveals domain specific self-regulated and active learning in biology. *The 2026 Annual meeting of American Educational Research Association*, Los Angeles, CA, United States.

Sung, H. (2026). Modeling multimodal interactions using epistemic network analysis: Key Considerations. In: Carmona, G., Lima, C., Santos, M.J., Benítez, H., Montero-Moguel, L., Galarza-Tohen, B. (eds) *Advances in Quantitative Ethnography. ICQE 2025. Communications in Computer and Information Science, vol 2677*. Springer, Cham. https://doi.org/10.1007/978-3-032-12229-2_19

Yu, L., **Sung, H.**, Plumley, R. D., Bernacki, M. L., & Greene, J. A. (2025). Exploring sequential patterns in academic help-seeking: Transitions and performance in a college science course. In Rajala, A., Cortez, A., Hofmann, R., Jornet, A., Lotz-Sisitka, H., & Markauskaite, L. (Eds.), *Proceedings of the 19th International Conference of the Learning Sciences - ICLS 2025* (pp. 1744-1748). International Society of the Learning Sciences. https://doi.org/10.22318/icls2025.563588

Yu, L., Garland, A., **Sung, H.**, Plumley, R. D., Greene, J. A., Fong, C. J., Halpin P. F., & Bernacki, M. L. (2025). Relationships among college students' academic help-seeking behaviors and performance in an Introductory science course. *The 2025 Annual meeting of American Educational Research Association*, Denver, CO, United States.

Yu, L., Ren, S., **Sung, H.**, Plumley, R. D., Garland, A., Greene, J. A., Fong, C. J., Halpin P. F., & Bernacki, M. L. (2025). Identifying college students' academic help-seeking groups and their relationships with antecedents and academic performance. *The 2025 Annual meeting of American Educational Research Association*, Denver, CO, United States.

Ha, J., Ku, D. H., Gong, H. J., Kang, S., Lee, J., Lee, Y., Park, K., **Sung, H.**, & Yu, S. (2025). Leveraging AI tools in teacher education: Bridging theory and practice for transformative learning. *The 2025 Annual meeting of Korean American Educational Research Association*, Denver, CO, United States.

Na, H. & **Sung, H.** (2024). The Effects of technology-enhanced embodied approach with augmented reality for geometry learning in K-12 Classrooms. *The 2024 Annual meeting of the Association for Educational Communications and Technology,* Kansas city, MO, United States.

Sung, H., Plumley, R. D., Bernacki, M. L., & Greene, J. A. (2024). Exploring differently motivated STEM learners' self-regulated learning processes. *American Psychological Association Annual Convention (Division 15)*, Seattle, WA, United States.

Sung, H., Berro, M., Yu, L., Plumley, R. D., Bernacki, M. L., & Greene, J. A. (2024). Exploring the relationship between students' adherence to active learning and self-regulated learning processes. In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), *Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024* (pp. 2361-2362). International Society of the Learning Sciences. https://doi.org/10.22318/icls2024.111230

Na, H. & **Sung, H.** (2024). The Effects of technology-enhanced embodied learning intervention for mathematics in K-12 classrooms. In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), *Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024* (pp. 2385-2386). International Society of the Learning Sciences. https://doi.org/10.22318/icls2024.161940

Sung, H., Xia, F., & Nathan, M. J. (2024). Using learners' machine-detected body movements to reveal their verbal and nonverbal contributions to the co-creation of embodied math knowledge. Presented at the round table session of *The 2024 Annual meeting of Korean American Educational Research Association*, Philadelphia, PA, United States.

Sung, H., Kim, D., Swart, M., & Nathan, M. (2023). Multimodal behavior analysis: Two patterns of collaborative construction of embodied knowledge. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45, Sydney, Australia. Retrieved from https://escholarship.org/uc/item/3nb4h58t

Grondin, M., **Sung, H.**, Dey I., & Luu, R. (2022). Positionality matters: Diversifying research teams can affect data analyses. Paper presented at *the Learning Sciences Graduate Students Conference*, Bloomington, IL, United States.

Sung, H., & Na, H. (2022). Learning math with gestures: Exploratory research of integrating a technology-enhanced embodied intervention into classrooms. Paper presented at *the 2022 Annual meeting of the Association for Educational Communications and Technology*, Las Vegas, NV, United States.

Sung, H., & Nathan, M. J. (2022). Unraveling temporally entangled multimodal interactions in CSCL environments. In Damşa, C. and Barany, A. (Eds.), *The Fourth International Conference on Quantitative Ethnography: Conference Proceedings Supplement* (pp. 163-165). Copenhagen, Denmark: The International Society for Quantitative Ethnography. https://www.qesoc.org/images/pdf/ICQE22_proceedings_supplement.pdf

Sung, H., Swart, M. I., & Nathan, M. J. (2022). Methods for analyzing temporally entangled multimodal data. In Weinberger, A. Chen, W., Hernández-Leo, D., & Chen, B. (Eds.), *Proceedings of the 15th International Conference on Computer-Supported Collaborative Learning – CSCL 2022* (pp. 242-249). Virtual: International Society of the Learning Sciences. https://doi.dx.org/10.22318/cscl2022.242 **[Awarded CSCL Naomi Miyake Best Student Paper]**

Sung, H., Swart, M., & Nathan, M. J. (2022). Teaching teachers teaching students: How embodied cognition can help pre-service teachers assess students' mathematical thinking. Paper presented at *the 2022 Annual meeting of American Educational Research Association*, San Diego, CA, United States.

Sung, H., & Nathan, M. J. (2021). Improving teachers' embodied mathematical understanding of students' gestures through online embodied learning activities. In *Proceedings of the Learning Sciences Graduate Students Conference* (pp. 5-6), Champaign, IL, United States.

Nachtigall, V., Nößler, A., & **Sung, H.** (2021). An epistemic network analysis of students' beliefs about natural and educational scientists. In Wasson, B., Zörgő, S. (Eds.), *Advances in Quantitative Ethnography. Communications in Computer and Information Science, Volume 1522.* Springer, Cham, Virtual. https://doi.org/10.1007/978-3-030-93859-8_14

Sung, H., Swart, M., & Nathan, M. J. (2021). Enhancing K-12 pre-service teachers' embodied understanding of the geometry knowledge through online collaborative design. In Olanoff, D., Johnson, K., & Spitzer, S. M. (Eds.), *Proceedings of the 43rd annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (pp. 909-917). Philadelphia, PA, United States.

Swart, M., Kirankumar, V., **Sung, H**., Xia, F., Kim, D., Kwon, O., Walkington, C., Schenck, K., & Nathan, M. J. (2021). Embodied transmission of ideas: Mathematical thinking through collaborative construction of geometry video game content. *Proceedings of the 43rd Annual Meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (pp. 1341-1345). Philadelphia, PA, United States.

Kirankumar, V., **Sung, H**., Swart, M., Kim, D., Xia, F., Kwon, O., Walkington, C., & Nathan, M. J. (2021). Embodied transmission of ideas: Collaborative construction of geometric content and mathematical thinking. In Hmelo-Silver, C. E., De Wever, B., & Oshima, J. (Eds.), *Proceedings of the 14th International Conference on Computer-Supported Collaborative Learning – CSCL 2021* (pp. 177-180). Virtual: International Society of the Learning Sciences. https://doi.dx.org/10.22318/cscl2021.177

Nachtigall, V., **Sung, H.** (2019). Students' collaboration patterns in a productive failure setting: An epistemic network analysis of contrasting cases. In: Eagan, B., Misfeldt, M., Siebert-Evenstone, A. (Eds.), *Advances in Quantitative Ethnography. ICQE 2019. Communications in Computer and Information Science, vol 1112.* Springer, Cham, Madison, WI, United States. https://doi.org/10.1007/978-3-030-33232-7_14

Sung, H., Cao, S., Ruis, A., & Shaffer, D. (2019). Reading for breadth, reading for depth: Understanding the relationship between reading and complex thinking using epistemic network analysis. In Lund, K., Niccolai, G. P., Lavoué, E., Hmelo-Silver, C., Gweon, G., & Baker, M. (Eds.), *Proceedings of the 13th International Conference on Computer Supported Collaborative Learning – CSCL 2019, Volume 1* (pp. 376-383). Lyon, France: International Society of the Learning Sciences. https://doi.dx.org/10.22318/cscl2019.376

Wu, B., **Sung, H.**, & Shaffer, D. W. (2019). Exploring the epistemic development trajectories in jigsaw-based collaborative problem solving. Paper presented at *the 2019 Annual Meeting of the American Educational Research Association*, Toronto, Canada.

Sung, H., Cao, T., Ruis, A. R., & Shaffer, D. W. (2019). An epistemic network analysis to investigate the interaction between reading and complex thinking using multimodal data. Poster presented at *the University of Wisconsin-Madison Education Research Poster Fair*, Madison, WI, United States.

Bhatt, H., Cao, T., **Sung, H.**, Siebert-Evenstone, A., Eagan, B., & Shaffer, D. W. (2019). Analysis of multi-modal learning data using horizon of observation ENA. Poster presented at *the University of Wisconsin-Madison Education Research Poster Fair*, Madison, WI, United States.

Sung, H., & Shaffer, D. W. (2018). An exploratory epistemic network analysis of combining multimodal data streams to understand student learning process. In *Proceedings of the Learning Sciences Graduate Students Conference* (pp. 172-173), Nashville, TN, United States.

Sung, H., Swiecki, Z., Wu, B., & Shaffer, D. W. (2018). Reading for breadth, reading depth: A multimodal analysis of student learning processes with epistemic network analysis. Poster presented at *the University of Wisconsin-Madison Education Research Poster Fair*, Madison, WI, United States.

Lee, H., **Sung, H.**, Park, Y., & Jo, I. (2015). Investigation influences of participation, regularity and centrality on learning achievement. Poster presented at *the 3rd Dubai International Conference in Higher Education*, Dubai, United Arab Emirates **[Awarded Highly Commended Poster Presentation]**

Lee, H., **Sung, H.**, Yu, J., Park, Y., & Jo, I. (2015). Clustering of online students: Towards an elaborated prediction model of learning achievement. Paper presented at *E-learning Korea 2015*, Seoul, Korea.

Lee, H., **Sung, H.**, Yu, J., Park, Y., & Jo, I. (2015). Development of prediction models based on the clustered online learners' behavioral patterns. In *Proceedings of The International Academic Conference on Teaching, Learning and E-learning (IAC-TLEI)*, Budapest, Hungary. ISBN: 978-80-905791-4-9

Sung, H., Lee, H., Park, Y. & Jo, I. (2014). Mediating effects of team activity on the relationship between students' centrality and learning achievement in online discussion-based class. *Proceedings of The International Conference of Educational Technology, Volume 2*. Seoul, Korea: The Koreans Society for Educational Technology.

Yoo, Y., Jo, I., Park, Y., Lee, H., & **Sung, H.** (2014). Analyzing online discussion activity: Using social network analysis with learning analytics approach. Paper presented at *the 2014 Annual meeting of the Association for Educational Communications and Technology Conference*, Jacksonville, FL, United States.

Kim, J., **Sung, H.**, Park, Y., & Jo, I. (2014). Applying learning analytics dashboard as a learning supporting tool: A case study of online statistics class. Paper presented at *E-learning Korea 2014*, Seoul, Korea. **[Awarded Gold Prize in Best Paper]**

Sung, H., Yoon, M., Park, Y., & Jo, I. (2014). Relationship of students' online behavior pattern and their psychological characteristics. Paper presented at *the 12th International Conference for Media in Education*, Seoul, Korea

INVITED TALKS AND PRESENTATIONS

Sung, H. (2024). The state of art in multimodal learning analytics. The 2nd Learning Sciences Research Institute, organized by Seoul National University, Seoul, Korea.

Sung, H. (2024). ChatGPT4o and multimodal learning analytics. 2024 LAPA Summer Symposium, organized by Ewha Womans University, Seoul, Korea.

Sung, H. (2022). Coding Nonverbal data *QE Sandbox*, Virtual. Retrieved from osf.io/m2x6g

Sung, H. (2022). Multimodal Learning Analytics. *Birds of a Feather (BOF) Networking session at Learning Analytics and Knowledge (LAK) conference*, Virtual.

Nathan, M. J., Swart, M., Kim, D., **Sung, H.**, & Xia, F. (2021). Instructional gestures: Handy ways to promote comprehension and learning. *The teaching and learning forum, organized by College and Steenbock libraries at University of Wisconsin-Madison*, Virtual.

Siebert-Evenstone, A., Swiecki, Z., Eagan, B., **Sung, H.**, & Shaffer, D. W. (2018). Developing and validating automated discourse codes: An introduction to nCodeR. In *Proceedings of the Learning Sciences Graduate Students Conference* (pp. 195-196), Nashville, TN, United States.

RESEARCH EXPERIENCE

Aug 2023–
Jul 2024

Postdoctoral fellow
Transformative Undergraduate Self-regulated STEM Learning and Education Research (NSF# 1920756)
University of North Carolina at Chapel Hill

- *PI: Bernacki, Matthew L. / Co-PI: Greene, Jeffrey A.*
- Investigated the relationship between students' think-aloud verbal data, behavioral log data, and academic success
- Applied epistemic network analysis to multimodal data to understand students' self-regulated learning processes

Jan 2023–
May 2023

Project Assistant
Designing Cross-curricular Integrated Mathematics with Social Studies
University of Wisconsin–Madison

- *PI: Agarwal, Priyanka*
- Led professional development program for teachers to design diversity and inclusion-focused cross-curricular math instruction
- Conducted interviews and qualitative analysis

Sep 2020–
May 2023

Research Assistant
Mathematical Action and Gesture in Instruction and Cognition Lab
University of Wisconsin–Madison

- *Advisor: Nathan, Mitchell J.*
- Led design and development of an online, collaborative embodied learning intervention for pre-service math teachers
- Orchestrated the entire project, including IRB, recruitment, data collection, data analysis, and research findings dissemination

Jul 2019–
Aug 2020

Project Assistant
Learning Internal Visualization Skills for Complex Engineering Concepts in Active Learning Classes (NSF# 1933078)
University of Wisconsin–Madison

- *PI: Rau, Martina A.*
- Led design and development of an intelligent tutoring system (ITS) through close collaboration with a programmer and content experts
- Conducted lab studies independently, leading data collection, data analysis, and research findings dissemination

Aug 2017–
Jul 2019

Research Assistant
Assessing Complex Collaborative STEM Learning at Scale with Epistemic Network Analysis (NSF# 1661036)
University of Wisconsin–Madison

- *PI: Shaffer, David W.*
- Led research project on multimodal analysis, combining behavioral log data and chat data to understand the learning process

- Contributed to the development of rENA (Epistemic Network Analysis in R) for accurate modeling of multiple streams of data

Sep 2014–Aug 2016 **Research Assistant**
Korea Research Foundation & Brain Korea 21 Plus
Ewha Womans University

- *Advisor: Jo, Il-Hyun*
- Led research project on prediction modeling of learning achievement using student behavioral log data
- Conducted social network analysis on discussion data collected in a learning management system (LMS) and MOOC context

Jul 2014–Aug 2014 **Research Assistant**
Institute for Teaching and Learning Center, Ewha Womans University

- Supported the digitization of instructional contents
- Conducted lecture video content analysis

TEACHING EXPERIENCE

University of Tennessee, Knoxville

Jan 2026–May 2026 **Instructor**
Department of Theory and Practice in Teacher Education
- *STEM 695: Capstone in Educational Data Science* (Graduate, online)

Conducted lectures throughout the semester

Jan 2026–May 2026 **Instructor**
Department of Theory and Practice in Teacher Education
- *STEM 680: Foundations in Educational Data Science* (Graduate, online)

Conducted lectures throughout the semester

Aug 2025–Dec 2025 **Instructor**
Department of Theory and Practice in Teacher Education
- *STEM 680: Foundations in Educational Data Science* (Graduate, online)
- Conducted lectures throughout the semester
- Course evaluation score: 4.40/5.0

Jan 2025–May 2025 **Instructor**
Department of Theory and Practice in Teacher Education
- *STEM 680: Foundations in Educational Data Science* (Graduate, online)
- Conducted lectures throughout the semester
- Course evaluation score: 4.86/5.0

Aug 2024–
Dec 2024

Instructor
Department of Theory and Practice in Teacher Education
- *STEM 680: Foundations in Educational Data Science* (Graduate, online)
- Conducted lectures throughout the semester
- Course evaluation score: 4.9/5.0

University of Wisconsin-Madison

Sep 2022–
Dec 2022

Lecturer
Department of Educational Psychology
- *EP 301: How people learn* (Undergraduate, in-person)
- Conducted lectures throughout the semester
- Instructor effectiveness score: 4.77/5.0

Jun 2022–
Aug 2022

Teaching Assistant
Learning Analytics Master's program
- *EP 551: Quantitative Ethnography* (Graduate, online)
  EP 501: Thinking and Learning (Graduate, online)
- Assisted students in data analysis through one-on-one lab sessions and graded weekly assignments and the final project
- TA Effectiveness score: 4.58/5.0

Jun 2021–
Aug 2021

Teaching Assistant
Learning Analytics Master's program
- *Analytics EP 501: Thinking and Learning* (Graduate, online)
- Graded weekly assignments and final project
- TA Effectiveness score: 4.71/5.0

Aug 2020–
May 2021

Tech Teaching Assistant
Department of Computer Sciences
- *CS 240: Introduction to Discrete Mathematics*
  CS 220: Data Science Programming I
  CS 354: Machine Organization and Programming
  (Undergraduate, online)
- Assisted in creating digital learning contents and facilitating (a)synchronous discussions

Apr 2020

Guest lecturer
Department of Educational Psychology
- *EP 301: How people learn* (undergraduate, in person)
- Conducted a lecture on *Technology in Education*

Aug 2019–
May 2020

Teaching Assistant
Department of Educational Psychology
- *EP 301: How people learn* (Undergraduate, in person)
- Graded weekly assignments and final project
- TA Effectiveness score: 4.45/5.0

University of Alabama

Sep 2022

Guest lecturer

Instructional Technology Master program
- *INTE 534: Issues and Trends in Instructional Technology* (Graduate, online)
- Conducted a lecture on *Current trends and issues in Learning Analytics*

Ewha Womans University

Jun 2022 **Guest lecturer**
School of Education
- *Educational Technology program* (Undergraduate and Graduate, hybrid)
- Conducted a lecture on *Research trends in Learning Sciences and Learning Analytics*

ADVISING & MENTORING EXPERIENCE

Dissertation Committee

2025 **Omiya Sultana.** Doctoral student. Ph.D. Dissertation Proposal. UT, Knoxville

Mentoring

2026 **Seungju Kim.** Postdoctoral scholar. KAIST
2024–2025 **Jaesung Hur.** Doctoral student. Florida State University
2024–2025 **Hunhui Na.** Doctoral student. Florida State University
2024 **Ju Lim.** Doctoral student. UW-Madison
2023–2024 **Linyu Yu.** Doctoral student. UNC-Chapel Hill
2023–2024 **Heewoo Lee.** Doctoral student. UNC-Chapel Hill
2023–2024 **Slki Narae Kim.** Doctoral student. UNC-Chapel Hill
2023 **Saerok Park.** Doctoral student. University of Central Florida
2022–2023 **Quinn Mierlak.** Undergraduate student. UW-Madison
2021–2023 **Fangli Xia.** Doctoral student. UW-Madison
2021–2023 **Jaeyoon Choi.** Doctoral student. UW-Madison
2021–2023 **Jihyun Rho.** Doctoral student. UW-Madison
2021 **Veena Kirankumar.** Undergraduate Research Intern. UW-Madison
2021 **Seongho Choi.** Graduate student. Global Engagement office, UW-Madison
2021 **Hae Sol Park.** Graduate student. Global Engagement office, UW-Madison
2021 **Hyelin Park.** Graduate student. Global Engagement office, UW-Madison
2021 **Manuel Rueda.** Graduate student. Global Engagement office, UW-Madison
2021 **Ting (Sophia) Cheng.** Graduate student. Global Engagement office, UW-Madison
2020 **Samyu Iyer.** Undergraduate Research Intern. UW-Madison
2019 **Claudia Ramly.** Doctoral Student. UW-Madison

2019	**Xiaoyuan (Ethan) Zhong.** Undergraduate Research Intern. UW-Madison
2018–2019	**Travis Cao.** Undergraduate Research Intern. UW-Madison
2017–2018	**Yuqing (Vanessa) Wu.** Undergraduate Research Intern. UW-Madison

EDITORIAL SERVICE

Peer-reviewed Journal Reviewer

2025–present	Computers & Education: X Reality
2024–present	Review of Educational Research
2024–present	Journal of Science Education and Technology
2024–present	Cognition and Instruction
2023–present	Journal of Educational Psychology

Annual Conference Proposal Reviewer

2024–present	International Conference on Quantitative Ethnography (ICQE)
2022–present	Annual meeting of American Educational Research Association (AERA)
2021–present	International Society of Learning Sciences (ISLS)
2021–2023	Psychology of Mathematics Education-North American Chapter Conference (PME-NA)
2018–2022	Learning Sciences Graduate Student Conference (LSGSC)

PROFESSIONAL SERVICE

Conference Committee Roles

2026–present	**Chair of External Affairs** Korean Society of Learning Sciences
2026	**Review Committee Member** 2026 AERA Division C Graduate Student Poster Award
2026–present	**Program Committee Member** International Conference on Quantitative Ethnography (ICQE)
2024–present	**Membership Committee Member** The International Society of the Learning Sciences (ISLS)
2021–2022	**Communications Committee Member** Learning Sciences Graduate Student Conference (LSGSC)
2020	**Host Committee Member** Learning Sciences Graduate Student Conference (LSGSC)
2019	**Program Committee Member** International Conference on Quantitative Ethnography (ICQE)
2018–2019	**Submissions Committee Member** Learning Sciences Graduate Student Conference (LSGSC)

Scholarly Community Involvement

2026–present	**Board member**

	International Society for Quantitative Ethnography (ISQE)
2025	**Chair/Network Coordinator** Korean EduTech/Learning Sciences Research Network (KELS)
2024–present	**Early Career Scholar** Korean American Educational Research Association (KAERA)
2023–2024	**Event Committee Member/Board Member** Korean EduTech/Learning Sciences Research Network (KELS)
2020	**Peer Mentor** Global Engagement Office, UW-Madison
2018–2019	**President** Korean Students and Scholars Association (KSSA), UW-Madison
2014–2017	**Executive Member** Korean Society for Educational Technology
2014–2017	**Executive Member** Association for Educational Communication and Technology
2014–2017	**Assistant Administrator** Korean Society for Learning and Performance

SKILLS AND CERTIFICATION

Research Methods and Software

- Specialized in multimodal data collection and analysis, including automated body movement detection, gesture, behavioral log data, speech, and physiological signals
- Expert in discourse analysis methods, particularly Epistemic Network Analysis (ENA)
- Advanced expertise in natural language processing (NLP) techniques
- Proficient in quantitative data analysis and related software (R, Python, SPSS)
- Proficient in qualitative data analysis and software tools (V-Note, Transana)
- Experience with JavaScript, AWS Transcribe (automatic speech recognition), and Cognitive Tutor Authoring Tools (CTAT)

Language

- English (fluent), Korean (native), Japanese (intermediate)

Certification

Teacher's license in Ethics (Certified by Korean Ministry of Education)